

Spiral Similarity

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1 Introduction

We are all familiar with the fundamental role angle chasing plays in Olympiad geometry. There are three major tools in angle chasing, namely:

- (i) Angle sum of triangles
- (ii) Similar triangles
- (iii) Concyclic points

Out of these three, (i) is the most basic and often its applications are most apparent. In fact, what a lot of Olympiad novices (including me when I first joined IMO training) do when they are attempting a geometry problem is to compute every angle on the diagram using angle sum of triangles.

Indeed, this approach often leads to nowhere, and that is where (ii) and (iii) (and indeed many other techniques) come in. However, these are much less obvious. Usually, it takes a lot of observation and even ingenuity to figure out a pair of similar triangles or cyclic quadrilaterals.

This is the reason why spiral similarity is a powerful tool. As we shall see, it has a close relationship with similar triangles and concyclic points. Thus, spiral similarity enables us to spot them quickly, making geometry problems substantially easier to solve.

2 Spiral Similarity

A geometric transformation is a function which sends the points on the plane to other points.

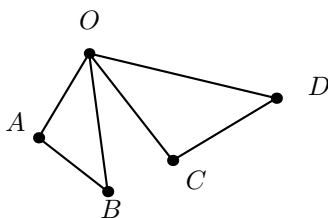
Definition 1 Suppose O is a point on the plane, and θ, r are numbers. Suppose A is a point on the plane, let A' be the point obtained by first rotating A through O by θ and dilating the image with respect to O by a factor of r .

The transformation sending A to A' is called the spiral similarity at point O with angle θ and ratio r . We say that this spiral similarity takes A to A' .

Remark 1. As you can convince yourself, it makes no difference if we dilate the point first then apply the rotation.

Remark 2. You will soon realize that the numbers θ and r are not important at all.

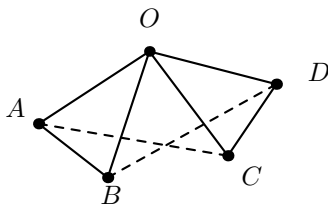
Theorem 1 Show that if a spiral similarity centered at O sends AB to CD , (which means it sends A to C and B to D), then $\triangle OAB$ and $\triangle OCD$ are directly similar.



Proof.

Remark 3. As promised, this shows that spiral similarity is closely related to similar triangles. It is actually a special case of similar triangles, with the same orientation and a point in common. These additional constraints allow us to deduce much more properties than general similar triangles.

Theorem 2 A spiral similarity with center O sends AB to CD if and only if it sends AC to BD .

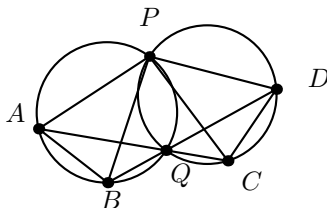


Proof.

Remark 4. You should start to see why spiral similarity is useful. Given a pair of similar triangles $\triangle OAB \sim \triangle OCD$, you automatically get another pair $\triangle OAC \sim \triangle OBD$. This allows us to spot similar triangles much quicker.

Remark 5. Indeed, in above the triangles should have the same orientation. However, this is not a strong condition at all. In the worse case scenario (the triangles are oppositely similar) one can simply reflect one of the triangles suitably to apply the above theorem. (This technique will be used in some of the exercise problems).

Theorem 3 Let A, B, C, D be points on the plane, suppose AB and CD intersect at Q and P is the second intersection of the circles (QAC) and (QBD) , then P is the centre of spiral similarity sending AC to BD .



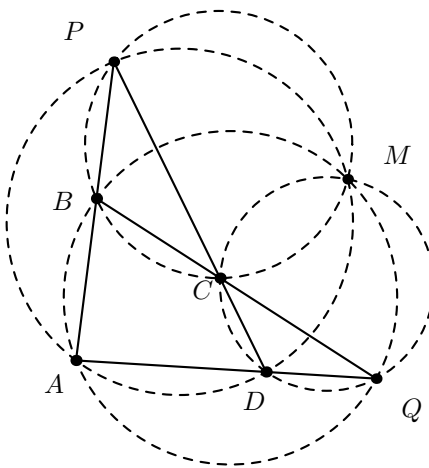
Proof.

Remark 6. This configuration appears extremely frequently in Olympiad Geometry. Given two circles intersecting at P and Q . From my experience:

- If there is a line passing through P and another passing through Q , one should immediately try to apply Reim's Theorem
- If there are two lines passing through Q , one should immediately try to apply the above theorem.

Combining Theorem 2 and Theorem 3 yields a striking result:

Theorem 4 (Miquel's Theorem) Suppose $ABCD$ is a quadrilateral, let P be the intersection of AB and CD , Q be the intersection of AD and BC , then the circles (PAD) , (PBC) , (QAB) and (QCD) are concurrent at a point M , called the Miquel Point of the quadrilateral.



Proof.

Remark 7. This theorem is very beautiful indeed. It echoes the last paragraph of the introduction and exhibits the relationship between spiral similarity and concyclic points. In (harder) problems considering the miquel points of some random quadrilateral will sometimes yield unexpected results.

3 Usage of Spiral Similarity

The space below is intentionally left blank. It is impossible for me to enumerate all uses of spiral similarity but you can write down anything you thought of in this space.

4 Example Problems

Try to finish them during our first lesson (29 October). You can ask me for hints if you want.

1. (CGMO 2006 P2) Let O be the intersection of the diagonals of convex quadrilateral $ABCD$. The circumcircles of $\triangle OAD$ and $\triangle OBC$ meet at O and M . Line OM meets the circumcircles of $\triangle OAB$ and $\triangle OCD$ at T and S respectively. Prove that M is the midpoint of ST .
2. (ELMO 2021 P1) In $\triangle ABC$, points P and Q lie on sides AB and AC , respectively, such that the circumcircle of $\triangle APQ$ is tangent to BC at D . Let E lie on side BC such that $BD = EC$. Line DP intersects the circumcircle of $\triangle CDQ$ again at X , and line DQ intersects the circumcircle of $\triangle BDP$ again at Y . Prove that D, E, X , and Y are concyclic.
3. (Geometry in Figures 4.1.6) Suppose $ABDC$ is a cyclic quadrilateral. Let X and Y be the foot of perpendicular from D to AB and BC . Let M and N be the midpoint of XY and AC respectively. Show that $DM \perp MN$.
4. (CMO 2010 P1) Two circles Γ_1 and Γ_2 meet at A and B . A line through B meets Γ_1 and Γ_2 again at C and D respectively. Another line through B meets Γ_1 and Γ_2 again at E and F respectively. Line CF meets Γ_1 and Γ_2 again at P and Q respectively. M and N are midpoints of arc PB and arc QB respectively. Show that if $CD = EF$, then C, F, M, N are concyclic.
5. (Korea MO Final Round 2018 P2) Triangle ABC satisfies $\angle ABC < \angle BCA < \angle CAB < 90^\circ$. O is the circumcenter of triangle ABC , and K is the reflection of O in BC . D, E is the foot of perpendicular line from K to line AB, AC , respectively. Line DE meets BC at P , and a circle with diameter AK meets the circumcircle of triangle ABC at $Q (\neq A)$. If PQ cuts the perpendicular bisector of BC at S , then prove that S lies on the circle with diameter AK .

5 Exercise Problems

We will do these on our second lesson (12 Nov). Indeed, if you are bored on the first lesson, you are 100% welcome to try these.

Remark 8. These problems may require tools beyond spiral similarity to solve (especially the hard ones).

Remark 9. Don't try to attempt all problems, set a reasonable target and try to accomplish that target.

5.1 Spiral Similarity

1. In $\triangle ABC$, P lies on BC and Q lies on the minor arc BC of (ABC) , furthermore suppose $\angle BAP = \angle CAQ$. Show that A is the centre of spiral similarity sending BP to QC .
2. In $\triangle ABC$, X is the centre of spiral sim. sending BA to AC . Show that
 - X lies on the A -symmedian of $\triangle ABC$
 - X, B, O, C are concyclic.
 - $XO \perp AX$
3. (Bulgarian MO 2019 P2) Let ABC be an acute triangle with orthocenter H and circumcenter O . Let the intersection points of the perpendicular bisector of CH with AC and BC be X and Y respectively. Lines XO and YO cut AB at P and Q respectively. If $XP + YQ = AB + XY$, determine $\angle OHC$.
4. (CMO 2020 P4) In acute triangle ABC ($AB > AC$), M is the midpoint of minor arc BC , O is the circumcentre of (ABC) and AK is its diameter. The line parallel to AM through O meets segment AB at D , and CA extended at E . Lines BM and CK meet at P , lines BK and CM meet at Q . Prove that $\angle OPB + \angle OEB = \angle OQC + \angle ODC$.
5. (USA TSTST 2012 P2) Let $ABCD$ be a quadrilateral with $AC = BD$. Diagonals AC and BD meet at P . Let ω_1 and O_1 denote the circumcircle and the circumcenter of triangle ABP . Let ω_2 and O_2 denote the circumcircle and circumcenter of triangle CDP . Segment BC meets ω_1 and ω_2 again at S and T (other than B and C), respectively. Let M and N be the midpoints of minor arcs $\widehat{SP}$ (not including B) and $\widehat{TP}$ (not including C). Prove that $MN \parallel O_1O_2$.
6. (IMO shortlist 2005 G5) Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocenter of triangle ABC , and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of triangle $\triangle ABC$ and triangle $\triangle ADE$.
7. (IMO shortlist 2018 G4) A point T is chosen inside a triangle ABC . Let A_1, B_1 , and C_1 be the reflections of T in BC, CA , and AB , respectively. Let Ω be the circumcircle

of the triangle $A_1B_1C_1$. The lines A_1T , B_1T , and C_1T meet Ω again at A_2 , B_2 , and C_2 , respectively. Prove that the lines AA_2 , BB_2 , and CC_2 are concurrent on Ω .

8. (IMO shortlist 2005 G4/Problem 5) Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and BC not parallel with DA . Let two variable points E and F lie of the sides BC and DA , respectively and satisfy $BE = DF$. The lines AC and BD meet at P , the lines BD and EF meet at Q , the lines EF and AC meet at R .

Prove that the circumcircles of the triangles PQR , as E and F vary, have a common point other than P .

9. (USATST 2018 P2) Let $ABCD$ be a convex cyclic quadrilateral which is not a kite, but whose diagonals are perpendicular and meet at H . Denote by M and N the midpoints of $\overline{BC}$ and $\overline{CD}$. Rays MH and NH meet $\overline{AD}$ and $\overline{AB}$ at S and T , respectively. Prove that there exists a point E , lying outside quadrilateral $ABCD$, such that

- ray EH bisects both angles $\angle BES$, $\angle TED$, and
- $\angle BEN = \angle MED$.

10. (IMO shortlist 2020 G5) Let $ABCD$ be a cyclic quadrilateral. Points K, L, M, N are chosen on AB, BC, CD, DA such that $KLMN$ is a rhombus with $KL \parallel AC$ and $LM \parallel BD$. Let $\omega_A, \omega_B, \omega_C, \omega_D$ be the incircles of $\triangle ANK, \triangle BKL, \triangle CLM, \triangle DMN$. Prove that the common internal tangents to ω_A , and ω_C and the common internal tangents to ω_B and ω_D are concurrent.
11. (CMO 2016 P2) In $\triangle AEF$, let B and D be on segments AE and AF respectively, and let ED and FB intersect at C . Define K, L, M, N on segments AB, BC, CD, DA such that $\frac{AK}{KB} = \frac{AD}{DC}$ and its cyclic equivalents. Let the incircle of $\triangle AEF$ touch AE, AF at S, T respectively; let the incircle of $\triangle CEF$ touch CE, CF at U, V respectively. Prove that K, L, M, N concyclic implies S, T, U, V concyclic.
12. (IMO shortlist 2018 G6/ Problem 6) A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point X lies inside $ABCD$ so that

$$\angle XAB = \angle XCD \quad \text{and} \quad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180^\circ$. Prove that the common internal tangents to ω_A , and ω_C and the common internal tangents to ω_B and ω_D are concurrent.

13. (IMO shortlist 2013 G6/ Problem 3) Let the excircle of triangle ABC opposite the vertex A be tangent to the side BC at the point A_1 . Define the points B_1 on CA and C_1 on AB analogously, using the excircles opposite B and C , respectively. Suppose that the circumcentre of triangle $A_1B_1C_1$ lies on the circumcircle of triangle ABC . Prove that triangle ABC is right-angled.
14. (Geometry in Figures 4.1.9) In $\triangle ABC$, O is the circumcenter. Let D and F be the foot of perpendicular from A to BC and C to AB . Let M be the midpoint of DF and E be a point on AM such that $OD \perp DE$. Show that $BE \perp AO$.
15. (Geometry in Figures 4.3.10) In $\triangle ABC$, the angle bisector of $\angle ABC$ and $\angle ACB$ meets the circumcircle of $\triangle ABC$ again at E and F . Let D be a point on the minor arc BC of (ABC) . Let X be the intersection of FD and BE , Y be the intersection of ED and CF . Let M be the midpoint of XY , show that FM and AY intersects on (ABC) .

16. (Iran TST 2022 Test 2 P3) Given a triangle ABC with circumcircle Γ . Points E and F are the foot of angle bisectors of B and C , I is incenter and K is the intersection of AI and EF . Suppose that T be the midpoint of arc BAC . Circle Γ intersects the A -median and circumcircle of AEF for the second time at X and S . Let S' be the reflection of S across AI and J be the second intersection of circumcircle of $AS'K$ and AX . Prove that quadrilateral $TJIX$ is cyclic.
17. (Iran TST 2017 Test 6 P6) In triangle ABC let O and H be the circumcenter and the orthocenter. The point P is the reflection of A with respect to OH . Assume that P is not on the same side of BC as A . Points E, F lie on AB, AC respectively such that $BE = PC$, $CF = PB$. Let K be the intersection point of AP, OH . Prove that $\angle EKF = 90^\circ$
18. (ELMO 2021 P6) In $\triangle ABC$, points D, E , and F lie on sides BC, CA , and AB , respectively, such that each of the quadrilaterals $AFDE$, $BDEF$, and $CEFD$ has an incircle. Prove that the inradius of $\triangle ABC$ is twice the inradius of $\triangle DEF$.

5.2 Miquel Point

1. Show that if $ABCD$ is a cyclic quadrilateral, P is the intersection of AB and CD , Q is the intersection of AD and BC , then the miquel point of $ABCD$ lies on PQ .
2. (EGMO 2022 P6) Let $ABCD$ be a cyclic quadrilateral with circumcenter O . Let the internal angle bisectors at A and B meet at X , the internal angle bisectors at B and C meet at Y , the internal angle bisectors at C and D meet at Z , and the internal angle bisectors at D and A meet at W . Further, let AC and BD meet at P . Suppose that the points X, Y, Z, W, O , and P are distinct. Prove that O, X, Y, Z, W lie on the same circle if and only if P, X, Y, Z , and W lie on the same circle.
3. (IGO 2021 Advanced P3) Consider a triangle ABC with altitudes AD, BE , and CF , and orthocenter H . Let the perpendicular line from H to EF intersects EF, AB and AC at P, T and L , respectively. Point K lies on the side BC such that $BD = KC$. Let ω be a circle that passes through H and P , that is tangent to AH . Prove that circumcircle of triangle ATL and ω are tangent, and KH passes through the tangency point.
4. (Iran Test 2016 Test 3 P2) Let ABC be an arbitrary triangle and O is the circumcenter of $\triangle ABC$. Points X, Y lie on AB, AC , respectively such that the reflection of BC WRT XY is tangent to circumcircle of $\triangle AXY$. Prove that the circumcircle of triangle AXY is tangent to circumcircle of triangle BOC .
5. (Iran Test 2017 Test 1 P5) In triangle ABC , arbitrary points P, Q lie on side BC such that $BP = CQ$ and P lies between B, Q . The circumcircle of triangle APQ intersects sides AB, AC at E, F respectively. The point T is the intersection of EP, FQ . Two lines passing through the midpoint of BC and parallel to AB and AC , intersect EP and FQ at points X, Y respectively. Prove that the circumcircle of triangle $TX Y$ and triangle APQ are tangent to each other.
6. (IMO shortlist 2020 G6) Let ABC be a triangle with $AB < AC$, incenter I , and A excenter I_A . The incircle meets BC at D . Define $E = AD \cap BI_A$, $F = AD \cap CI_A$. Show that the circumcircle of $\triangle AID$ and $\triangle I_AEF$ are tangent to each other

7. (China TST 2015 Test 1 Problem 3) Let $\triangle ABC$ be an acute triangle with circumcenter O and centroid G . Let D be the midpoint of BC and $E \in \odot(BC)$ be a point inside $\triangle ABC$ such that $AE \perp BC$. Let $F = EG \cap OD$ and K, L be the point lie on BC such that $FK \parallel OB, FL \parallel OC$. Let $M \in AB$ be a point such that $MK \perp BC$ and $N \in AC$ be a point such that $NL \perp BC$. Let ω be a circle tangent to OB, OC at B, C , respectively .

Prove that $\odot(AMN)$ is tangent to ω

8. (APMO 2014 P5) Circles ω and Ω meet at points A and B . Let M be the midpoint of the arc AB of circle ω (M lies inside Ω). A chord MP of circle ω intersects Ω at Q (Q lies inside ω). Let ℓ_P be the tangent line to ω at P , and let ℓ_Q be the tangent line to Ω at Q . Prove that the circumcircle of the triangle formed by the lines ℓ_P, ℓ_Q and AB is tangent to Ω .
9. (IMO shortlist 2020 G7) Let P be a point on the circumcircle of acute triangle ABC . Let D, E, F be the reflections of P in the A -midline, B -midline, and C -midline. Let ω be the circumcircle of the triangle formed by the perpendicular bisectors of AD, BE, CF . Show that the circumcircles of $\triangle ADP, \triangle BEP, \triangle CFP$, and ω share a common point.